DAY-04(23.07.2026)

TCP/IP MODELS:

layered networking framework(4-layered)

Transmission Control Protocol / Internet Protocol

1. Application layer
2. Transport layer
3. Internet layer
4. Network access

APPLICATION LAYER:

- applications interact with the networks
- protocols - HTTPS(Hyper Text Transfer Protocol), DNS(Domain Name System), FTP(File Transfer Protocol), SMTP(Simple Mail Transfer Protocol)
- data formatting, encryption

TRANSPORT LAYER:

- data transferred as packets(regulates data flow) → Sliding window protocol
- protocols - TCP, UDP
- port num - to allow multiple apps to share the network simultaneously

INTERNET LAYER:

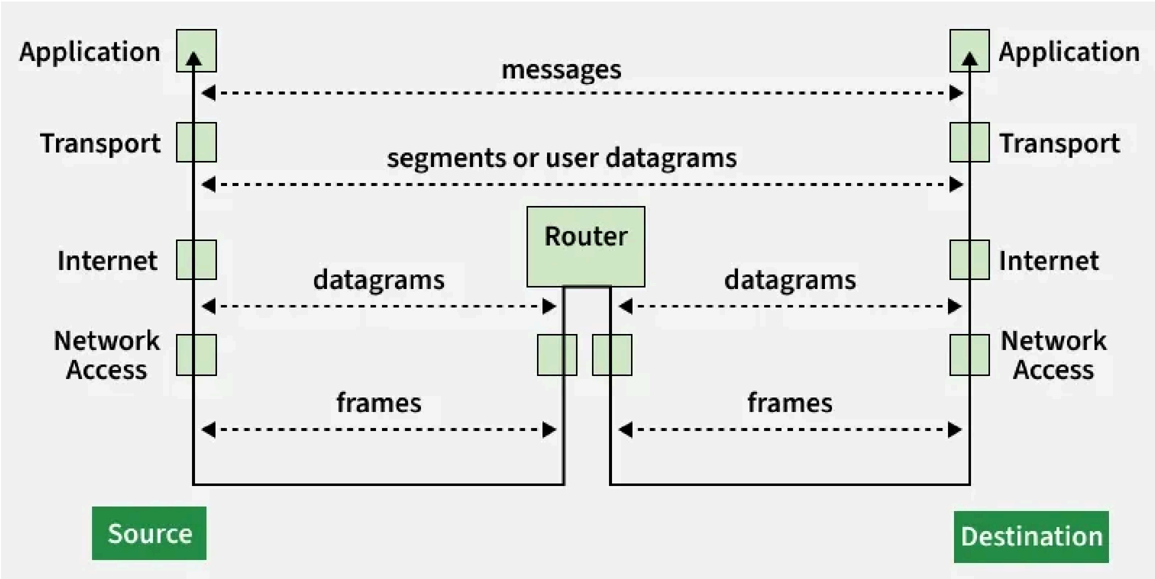
- addressing, packaging, routing the best
- ip address → best path →packets to small reassembles
- protocols used - IP(primary), ICMP[Internet Control Message Protocol](error reporting), ARP[Address Resolution Protocol](address resolution)

NETWORK ACCESS:

- transmitting data over network hardware
- organizes data into frames
- transmission errors(checksums/CRC)

- uses address →identification

▼ WORKING:



- Application → messages ←complete data
- Transport → segments ←reassembles
- Internet →packets ←unpack IP to reach correct
- Network Access → frames ← errors check ← unpacked

ADVANTAGES:

- widely used
- platform independent
- scalability
- reliable communication
- free to use(open standard)

DISADVANTAGES:

complex for beginners

less security needs TLS/SSL

limited multimedia support

overhead

▼

TCP[Transmission Control Protocol]:

- ensures data delivered completely
- detects error in data
- if any data is lost or corrupted it resends to the receiver
- data → packets
- it provides connection between the sender and receiver before the transmission of data
- ex : web page, files, emails

▼

UDP [User Datagram Protocol]:

- it is faster but do not ensure that the data is delivered completely or not.
- detects error but do not resends it or recover if the data is lost
- it avoids extra checking even though it is faster
- ex : live streaming, online gaming, real time